

Condensed Matter Physics

Lecture Notes

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Contents

Recommended Books and Resources	2
1 Tight Binding Model in One Dimension	3
1.1 Solution of the Tight Binding Chain	3
2 Quantum Many Body physics	5

Recommended Books and Resources

Primary references:

- Neil W. Ashcroft, N. David Mermin, *Solid State Physics*.
- Charles Kittel, *Introduction to Solid State Physics*.
- J. J. Sakurai, Jim Napolitano, *Modern Quantum Mechanics*.

1 Tight Binding Model in One Dimension

As we discussed in the class, consider a 1D chain of N atoms (periodic boundary conditions, site $N \equiv$ site 0), one orbital $|n\rangle$ per site, orthonormal:

$$\langle n|m\rangle = \delta_{n,m}. \quad (1)$$

Trial wavefunction:

$$|\Psi\rangle = \sum_n \phi_n |n\rangle, \quad (2)$$

giving the effective Schrödinger equation (Assingment for you guys!)

$$\sum_m H_{nm} \phi_m = E \phi_n, \quad H_{nm} = \langle n|H|m\rangle. \quad (3)$$

Calculation of Matrix elements

Writing $H = K + \sum_j V_j$ (kinetic + sum of atomic potentials), and treating $|m\rangle$ as the atomic orbital on site m :

$$H_{n,m} = \epsilon_{\text{atomic}} \delta_{n,m} + \sum_{j \neq m} \langle n|V_j|m\rangle. \quad (4)$$

The second term is nonzero only for nearby sites (electron “hopping”):

$$\sum_{j \neq m} \langle n|V_j|m\rangle = \begin{cases} V_0 & n = m \\ -t & n = m \pm 1 \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

Hence,

$$H_{n,m} = \epsilon_0 \delta_{n,m} - t(\delta_{n+1,m} + \delta_{n-1,m}), \quad \epsilon_0 \equiv \epsilon_{\text{atomic}} + V_0. \quad (6)$$

This defines the **tight-binding chain**: ϵ_0 is the on-site energy, t is the hopping amplitude.

1.1 Solution of the Tight Binding Chain

Try a Bloch-type ansatz

$$\phi_n = \frac{e^{ikna}}{\sqrt{N}}. \quad (7)$$

Substituting into Eq. (6) and using periodicity ($k \rightarrow k + 2\pi/a$ equivalent, k quantized in units of $2\pi/L$, $L = Na$):

$$E(k) = \epsilon_0 - 2t \cos(ka), \quad k \in \left[-\frac{\pi}{a}, \frac{\pi}{a}\right]. \quad (8)$$

- Bandwidth = $4|t|$; band runs from $\epsilon_0 - 2|t|$ to $\epsilon_0 + 2|t|$.
- Larger t (closer atoms) $\Rightarrow$ wider band.

Effective mass near band bottom

For small k (expand $\cos ka$):

$$E(k) \approx \text{const} + ta^2k^2. \quad (9)$$

Comparing to free-electron dispersion $E_{\text{free}}(k) = \frac{\hbar^2 k^2}{2m}$, define an **effective mass** m^* :

$$\frac{\hbar^2}{2m^*} = ta^2 \quad \implies \quad m^* = \frac{\hbar^2}{2ta^2}. \quad (10)$$

Near the band bottom, electrons behave like free particles of mass m^* — independent of the true electron mass, but set by the hopping t and lattice spacing a .

2 Quantum Many Body physics

I will update this section after discussing above topic in the class.